Tutorial - 11 (Confusion Matrix)

Dataset

Simple	Feature 1 (Age)	Feature 2 (Income)	Feature 3 (Credit Score)	Actual class	Predicted Probility
S1	25	40	650	1	0.90
S2	30	60	700	1	0.75
S3	45	35	600	0	0.60
S4	50	30	580	0	0.40
S5	28	55	720	1	0.30

Based on thershold : 0.5.

(Vi).

Actual class	Pred Prob	Pred Output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	0 (TN)
1	0.30	0 (FN).

Class 1 → Positive
class 0 → Negative.

1) Thershold : 0.5
Prob ≥ 0.5 → class 1
Prob < 0.5 → class 0.

Confusion Matrix

	Actual 1	Actual 0
Predicted 1	TP 2	FP 1
Predicted 0	FN 1	TN 1

2×2.

Precision :- $\frac{TP}{TP+FP} = \frac{2}{2+1} = \frac{2}{3} = 0.67$

9

• Recall. / (True Positive Rate) :-

$$= \frac{TP}{TP + FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$$

• F1- Score :-

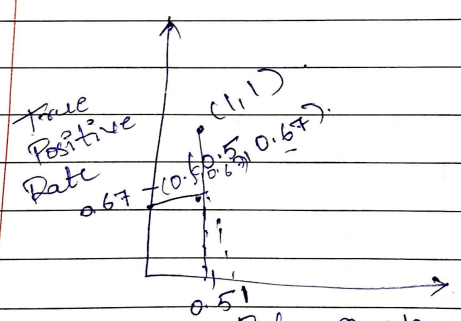
$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 0.67 \times 0.67}{0.67 + 0.67} = 0.67$$

2). False Positive Rate (FPR)

$$\frac{FP}{FP + TN} = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

3). ROC curve



= Area under curve

$= 0.67 \times 1$

$= 0.67$

Threshold : 0.7.

Y	Pred Prob	Pred output (0.7)	0.6	0.4	0.3
1	0.90	1	1	1	1
1	0.75	1	1	1	1
0	0.60	0	1	1	1
0	0.40	0	0	1	1
1	0.30	0	0	0	1

$\geq 0.7 \rightarrow$ class 1
$< 0.7 \rightarrow$ class 1.

Confusion Matrix

0.7

	Actual (1)	Actual (0)
Pred (1)	TP 2	FP
Pred (0)	FN 1	TN 2

0.6

	TP 2	FP 1
	FN 1	TN 1

Actual	Pred	
0	0	- TN
1	1	- TP
0	1	- FP
1	0	- FN

$= \text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{2+1} = \frac{2}{3} = 0.67$

False

$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{2+0} = \frac{2}{2} = 1.$

2) False Positive Rate (FPR)

$= \frac{FP}{FP+TN} = \frac{0}{0+2} = \frac{0}{2} = 0$

$= 0.6.$

$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+1} = 0.67$

$(FPR) = \frac{FP}{FP+TN} = \frac{1}{1+1} = 0.5$

Quick Work

Page No.:

Date:

M T W T F S S

9

0.4

PH 1	TP 2.	FP 2.
PH 0	FN. 1	TN 0.
	AC 1	AC 0.

0.3

TP 3.	FP 1
FN 0.	TN 0.

$= \text{Recall} :- \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67.$

$= FPR = \frac{FP}{FP+TN} = \frac{2}{2+0} = 1.$

0.3

$\text{Recall} - \frac{TP}{TP+FN} = \frac{3}{3+0} = 1.$

$FPR = \frac{FP}{FP+TN} = \frac{1}{1+0} = 1.$